

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

GRAVITATION

LECTURE-1

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1 Universal Law of Gravitation ✓
- 2 Vector Form of Gravitational Force ✓
- 3 Acceleration due to Gravity/Gravitational Field ✓
- 4 Variation of (g) with Height
- 5 Variation of (g) with Depth

Gravitation

* Basic

* Kepler law

* universal law of gravitation

* $g \Rightarrow$ Gravitational field / all due to gravity

* variation of (g) (a) height
(b) depth

* Gravitational PE

* Escape velocity

* orbital velocity

Basic



Board



Neet / Jee

Gravitation

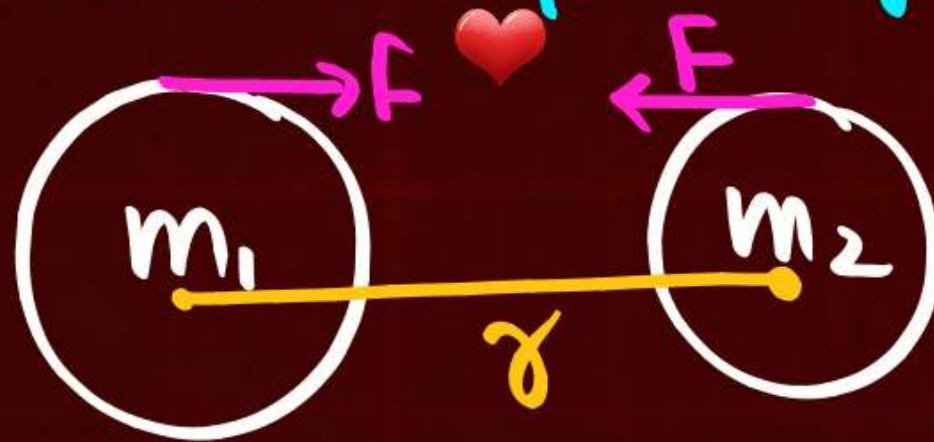
Kepler's laws of planetary motion, universal law of gravitation.
Acceleration due to gravity and its variation with altitude and depth.
Gravitational potential energy and gravitational potential, escape
speed, orbital velocity of a satellite.

Universal Law of Gravitation

(Newton)

Every mass attract every other mass with the force which is directly proportional to product of mass. and inversely proportional

to the square of distance b/w them



$$\uparrow F \propto m_1 m_2 \uparrow$$

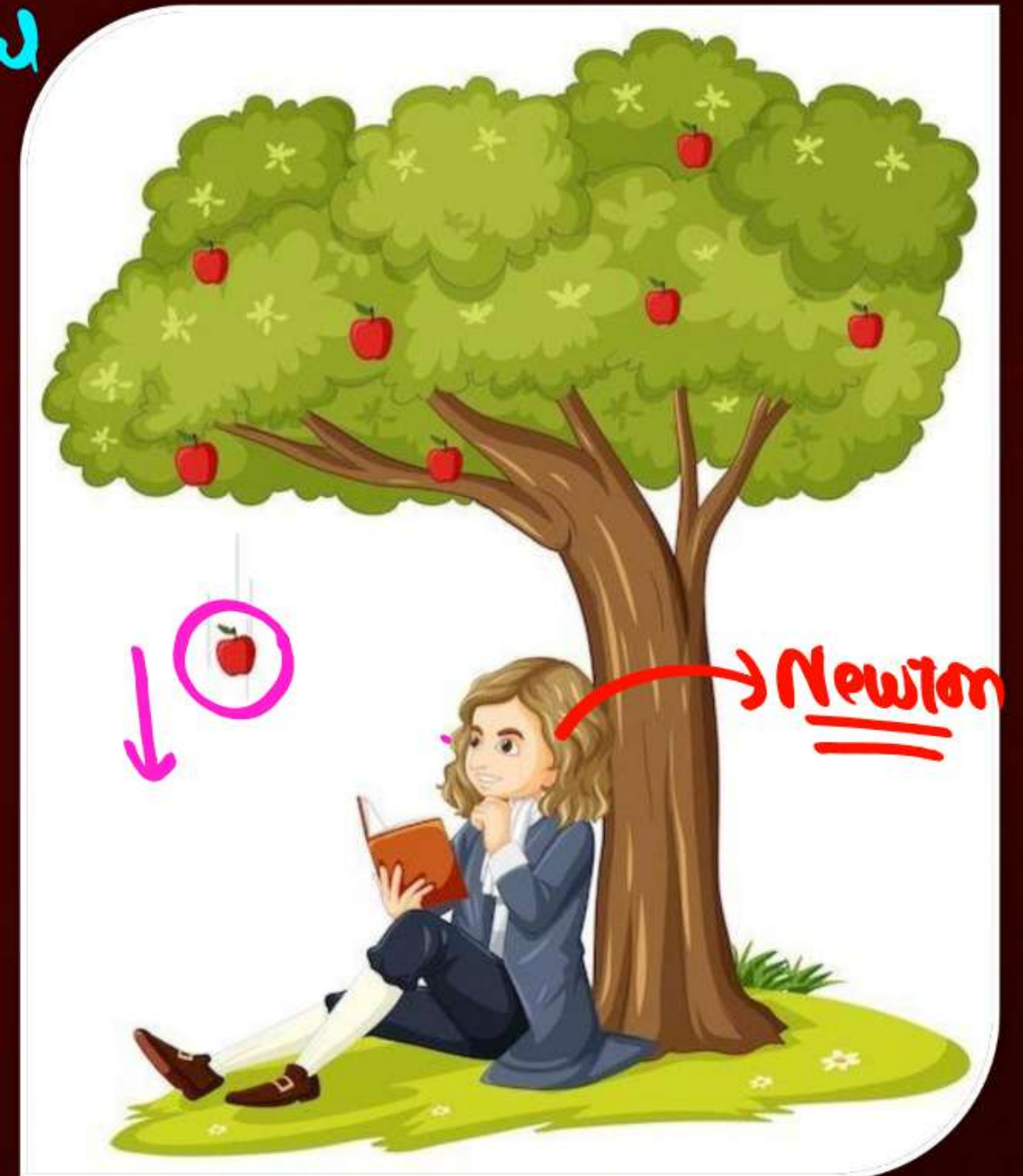
$$\downarrow F \propto \frac{1}{r^2} \uparrow$$

$$F \propto \frac{m_1 m_2}{r^2}$$

$$F = G \frac{m_1 m_2}{r^2}$$

$G \Rightarrow$ Gravitational Constant

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$



Unit of (G) (Gravitational constant)

$$F = \frac{G m_1 m_2}{r^2}$$

$$F r^2 = G m_1 m_2$$

$$\frac{F r^2}{m_1 m_2} = G$$

$$\frac{N m^2}{kg \cdot kg} = G$$

$$\frac{N m^2}{kg^2} = G$$

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Dimensional formula of G = $M^{-1} L^3 T^{-2}$

Vector Form of Gravitational Force

- Newton Give the magnitude of Gravitational force

$$F = G \frac{m_1 m_2}{r^2}$$

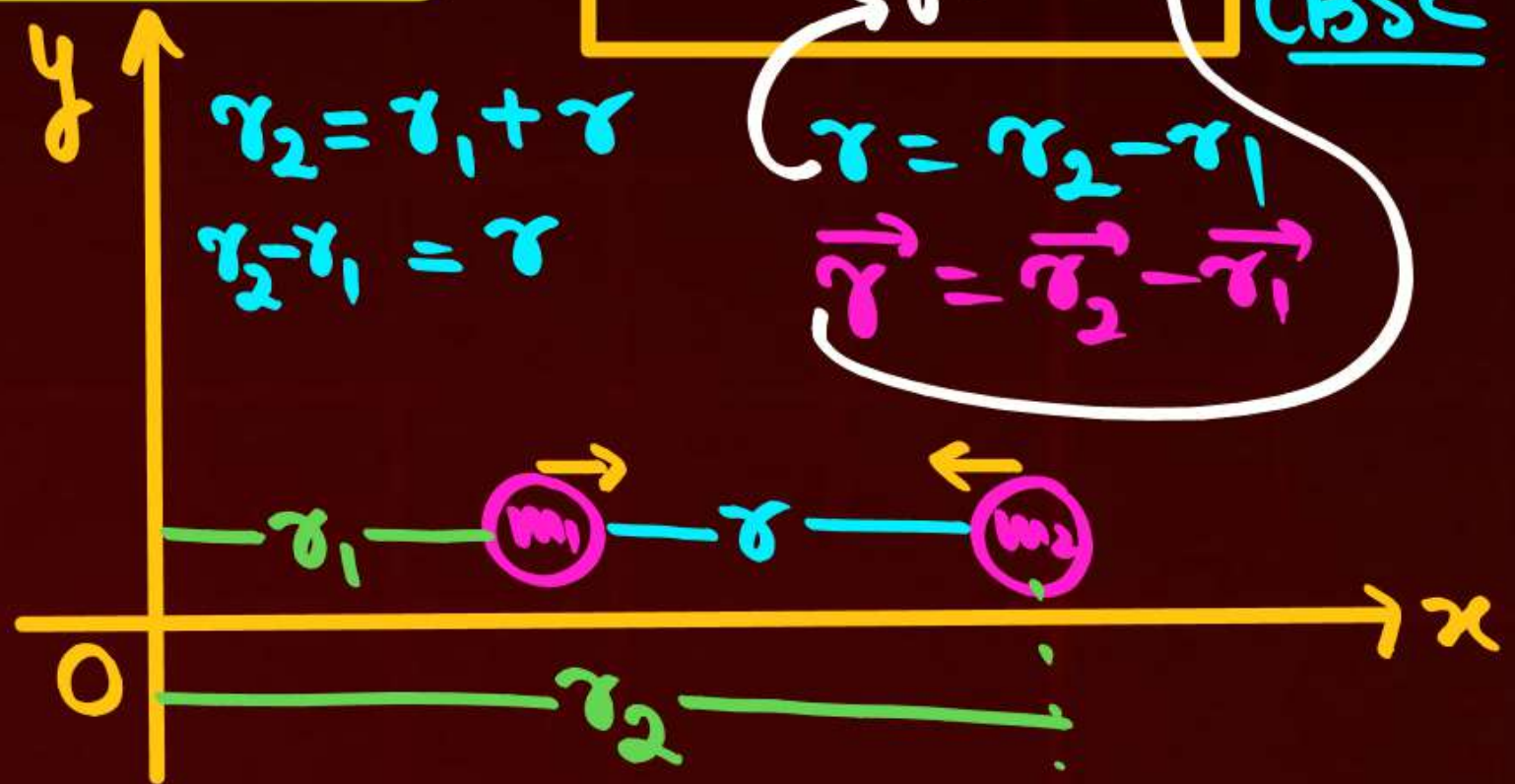
Vector = magnitude $\times$ Direction $\hat{n}$

$$\vec{F} = G \frac{m_1 m_2}{r^2} \times \hat{n} \quad \hat{n} = \frac{\vec{r}}{r}$$

$$\vec{F} = G \frac{m_1 m_2}{r^2} \frac{\vec{r}}{r}$$

$$\vec{F} = G \frac{m_1 m_2}{r^3} \vec{r}$$

CBSC



$$F = G \frac{m_1 m_2}{(r_2 - r_1)^3} (\vec{r}_2 - \vec{r}_1)$$

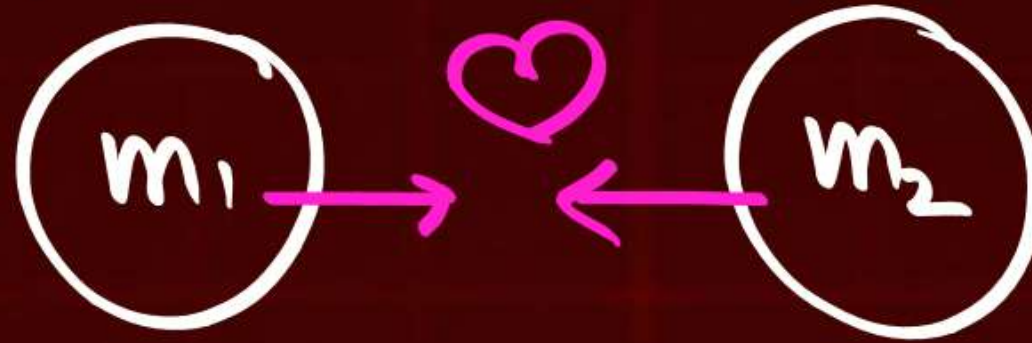
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Note - (I)

Gravitational Force

- ✓ The force of attraction b/w any two bodies in a universe



Note - (II)

Gravity

- ✓ The force of attraction b/w earth and any body is called Gravity



Can you solve it?

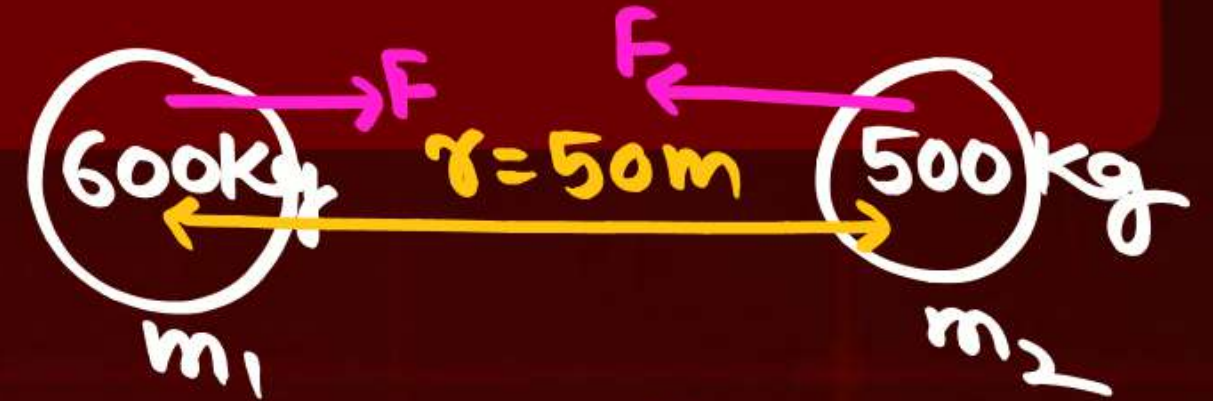


Spheres with weight 600 and 500 kg are kept at a distance 50 m apart. If the value of G is 6.6×10^{-11} unit then find the force of attraction between them.

Sol $F = \frac{G m_1 m_2}{r^2}$

$$F = \frac{6.67 \times 10^{-11} \times 600 \times 500}{(50)^2}$$

$$F = \frac{6.67 \times 10^{-11} \times 600 \times 500}{50 \times 50}$$



$$F = \frac{6 \times 600 \times 10^{-11}}{5}$$

$$F = \frac{6 \times 120 \times 10^{-11}}{5}$$

$$F = 6 \times 120 \times 10^{-11}$$

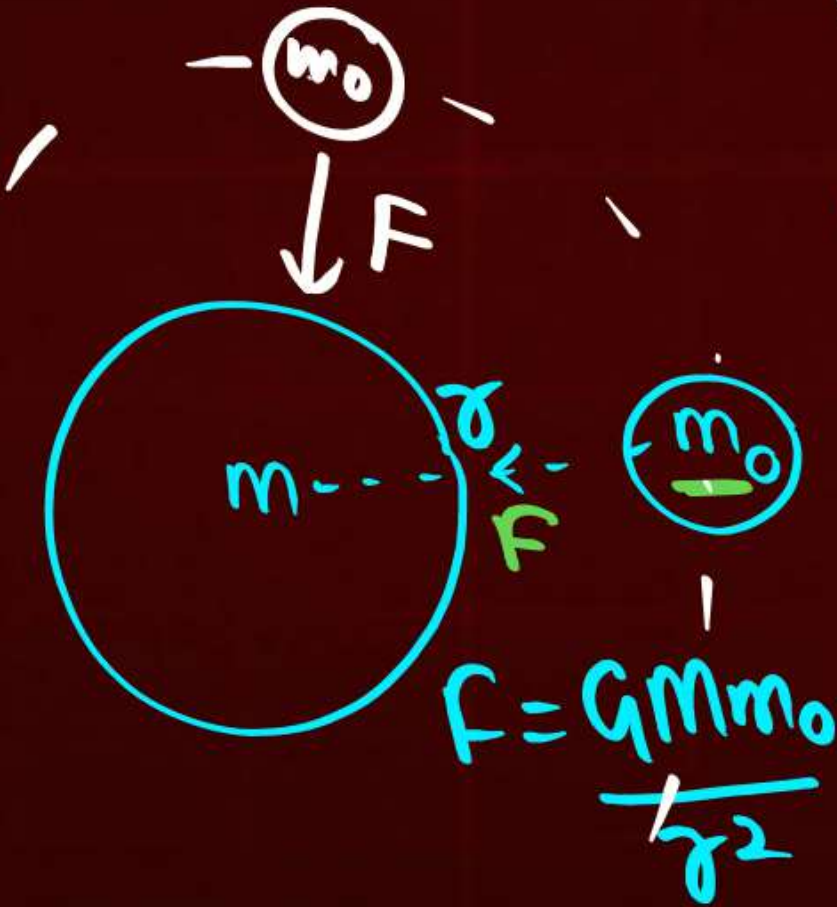
$$F = 720 \times 10^{-11} \text{ N} \checkmark$$

Acceleration due to Gravity/Gravitational Field

Gravitational field is the area around the mass where force of attraction can be experienced.

Or

Gravitational field is force acting per unit Mass (m_0)



$$g = \frac{F}{m_0}$$

$$g = \frac{GMm_0}{r^2}$$

$$g = \frac{GM \cancel{m_0}}{r^2 \cancel{m_0}} = \frac{GM}{r^2}$$



Another formula of (g) (in terms of Density) f

$$\# g = \frac{Gm}{r^2} \quad \text{--- (I)}$$

$$g = \frac{G \left(\frac{4}{3} \pi r^3 \rho \right)}{r^2}$$

$$g = G \frac{4}{3} \pi r \rho \quad \text{--- (II)}$$



$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$\text{Mass} = m$$

$$\text{Density} = \frac{m}{V}$$

$$\rho = \frac{m}{\frac{4}{3} \pi r^3}$$

$$\frac{4}{3} \pi r^3 \rho = \underline{\text{Mass}}$$

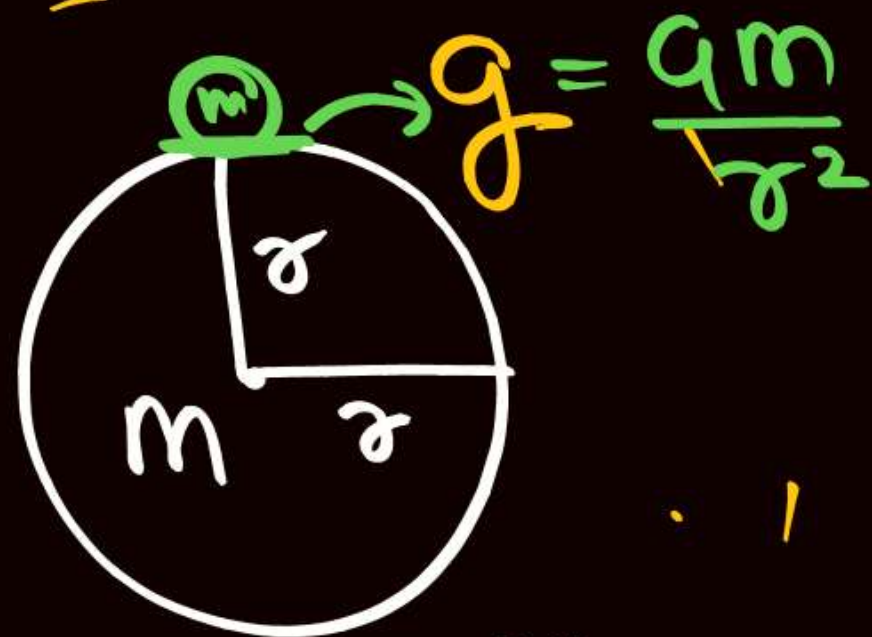
Agga Jaa k use karina hai.

Acceleration due to gravity of a earth

$$g = 10 \text{ m/s}^2$$

$$g = 9.8 \text{ m/s}^2$$

Proof



$$m = 6 \times 10^{24} \text{ kg}$$

$$r = 6 \times 10^6 \text{ m}$$

$$\# g = \frac{Gm}{r^2} \quad G = 6.67 \times 10^{-11}$$

$$g = \frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{(6 \times 10^6)^2}$$

$$g = \frac{6 \times 6 \times 10^{-11} \times 10^{24}}{6 \times 6 \times 10^6 \times 10^6}$$

$$g = \frac{10^{-11+24}}{10^{6+6}}$$

$$g = \frac{10^{13}}{10^{12}}$$

$$g = 10 \text{ m/s}^2$$

$$g = 9.8 \text{ m/s}^2$$

$$\downarrow g = -10 \text{ m/s}^2 = -9.8 \text{ m/s}^2$$





Variation of (g)

$$g = -10 \text{ m/s}^2$$

$$\uparrow \downarrow \downarrow \uparrow \quad g = \frac{um \uparrow \downarrow}{r^2 \uparrow \downarrow}$$

$$(i) \quad r \uparrow \quad g \downarrow$$

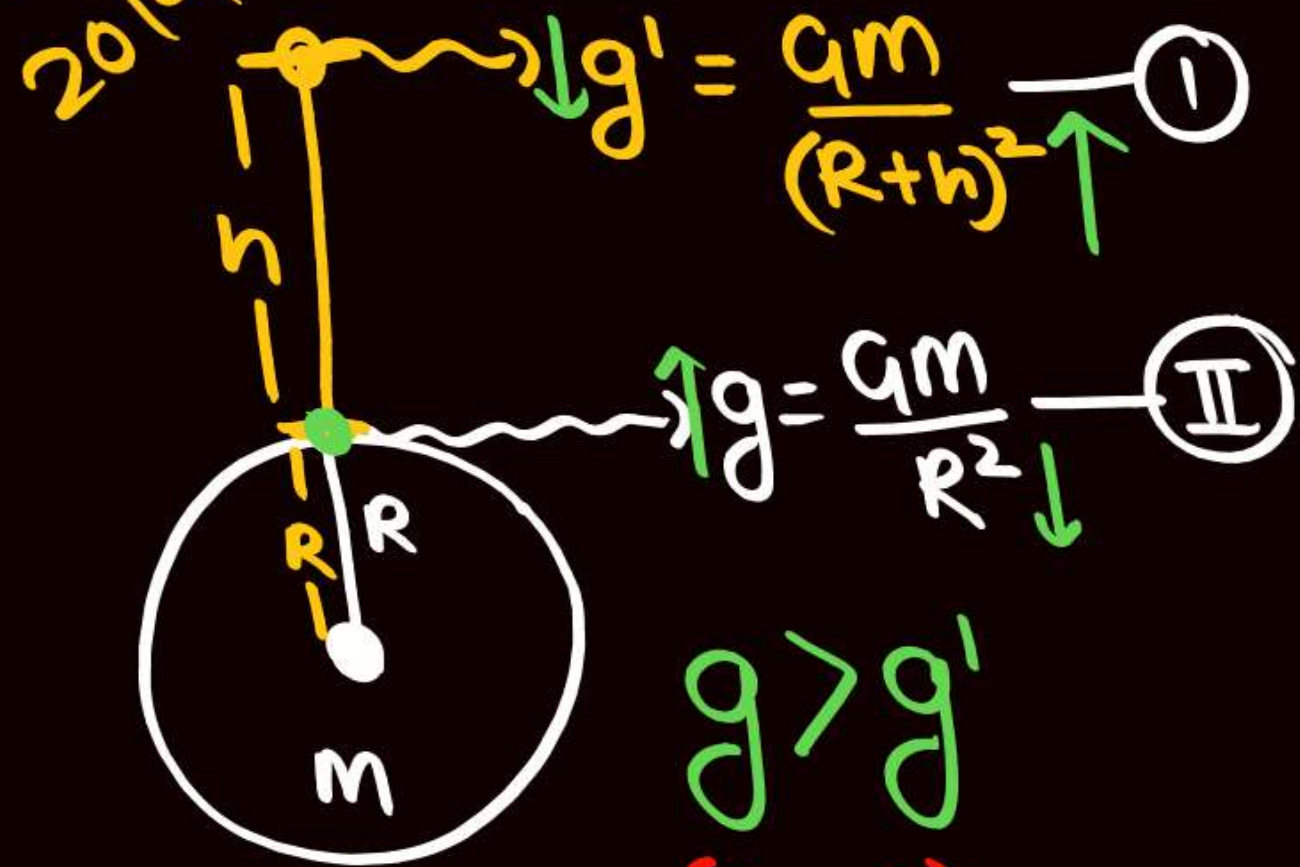
$$(ii) \quad m \uparrow \quad g \uparrow$$

Note

our (g) is not constant

(i) Variation of g with height

let us consider our
Earth of mass (m)
and Radius (R)



$$g > g'$$

(m.c)

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Divide (I) and (II)

$$\frac{g'}{g} = \frac{\frac{GM}{(R+h)^2}}{\frac{GM}{R^2}} \quad \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \times \frac{d}{c}$$

$$\frac{g'}{g} = \frac{\cancel{GM}}{(R+h)^2} \times \frac{R^2}{\cancel{GM}}$$

$$\frac{g'}{g} = \frac{R^2}{(R+h)^2}$$

$$\frac{g'}{g} = \frac{R^2}{(\underline{R} + \frac{h\underline{R}}{R})^2}$$

$$\frac{g'}{g} = \frac{R^2}{\left(R\left(1 + \frac{h}{R}\right)\right)^2}$$

$$\frac{g'}{g} = \frac{\cancel{R^2}}{\cancel{R^2}\left(1 + \frac{h}{R}\right)^2}$$

$$\frac{g'}{g} = \frac{1}{\left(1 + \frac{h}{R}\right)^2}$$

$$\frac{g'}{g} = \left(1 + \frac{h}{R}\right)^{-2}$$

$$\frac{g'}{g} = \left(1 + -2\frac{h}{R}\right)$$

$$\frac{g'}{g} = \left(1 - 2\frac{h}{R}\right)$$

$$(1+x)^n = 1+nx$$

Binomial
theorem

$$g' = g\left(1 - 2\frac{h}{R}\right)$$

numerical

$$\underline{g=10}$$

$$g' = g \left(1 - \frac{2h}{R}\right)$$

$$\underbrace{g'} = \underbrace{g} - \underbrace{g \frac{2h}{R}}$$

$$g' < \textcircled{g}$$

$$g' = 10 - 2$$

$$g' = 8$$

EXPECTATION



Chalo Solve krte hai?

Q/W



Find the value of g at a height of 10 km from the surface of earth.
One earth's surface $g = 9.8 \text{ m/s}^2$ and radius of earth is 6400 km.

$$g = \underline{9.8 \text{ m/s}^2}$$

10 m/s²

$$g' = g \left(1 - \frac{2h}{R} \right)$$

$$h = 10 \text{ km}$$

$$= 10 \times 1000 \text{ m}$$

$$= \underline{10000 \text{ m}}$$

$$R = 6400 \text{ km} = \underline{6 \times 10^6 \text{ m}}$$

$$= 6400 \times 1000 \text{ m} = 6400000 \text{ m} \checkmark = \underline{6.4 \times 10^6 \text{ m}}$$

SUMMARY



$$\textcircled{1} F = G \frac{m_1 m_2}{r^2}$$

$$\textcircled{2} \textcircled{g}$$

$$\textcircled{3} g' = g \left(1 - \frac{2h}{R} \right)$$

"BELIEVE IN YOURSELF"

"EVEN WHEN NO ONE

ELSE DOES."

Thank
You

